

This LM is to add security group module and use it in the parent main.tf

Steps

1. Create the security group module

In the module github, create an “**security-groups**” folder

Filename1: sg.tf

```
# Security Group - base security group (rules defined separately)
```

```
resource "aws_security_group" "web_sg" {  
  name      = "lab9-web-sg-${var.project}"  
  description = "Security group for web server"  
  vpc_id    = var.vpc_id
```

```
  tags = {  
    Name = "lab9-web-sg-${var.project}"  
  }  
}
```

```
# Security Group Rule - HTTP ingress
```

```
resource "aws_vpc_security_group_ingress_rule" "http" {  
  security_group_id = aws_security_group.web_sg.id  
  description       = "HTTP"  
  from_port         = 80  
  to_port           = 80  
  ip_protocol       = "tcp"  
  cidr_ipv4         = "0.0.0.0/0"
```

```
  tags = {  
    Name = "lab9-http-rule-${var.project}"  
  }  
}
```

```
# Security Group Rule - ssh ingress
```

```

resource "aws_vpc_security_group_ingress_rule" "ssh" {
  security_group_id = aws_security_group.web_sg.id
  description      = "ssh"
  from_port        = 22
  to_port          = 22
  ip_protocol       = "tcp"
  cidr_ipv4         = "0.0.0.0/0"

  tags = {
    Name = "lab9-http-rule-${var.project}"
  }
}

# Security Group Rule - Allow all outbound
resource "aws_vpc_security_group_egress_rule" "all_outbound" {
  security_group_id = aws_security_group.web_sg.id
  description       = "Allow all outbound"
  ip_protocol        = "-1"
  cidr_ipv4          = "0.0.0.0/0"

  tags = {
    Name = "lab9-egress-rule-${var.project}"
  }
}

```

Filename2: sg_var.tf

```

variable "vpc_id" {
  description = "VPC ID where security groups will be created"
  type        = string
}

```

```

variable "project" {
    description = "Project name - use UserX format where X is your student number"
    type        = string
}

```

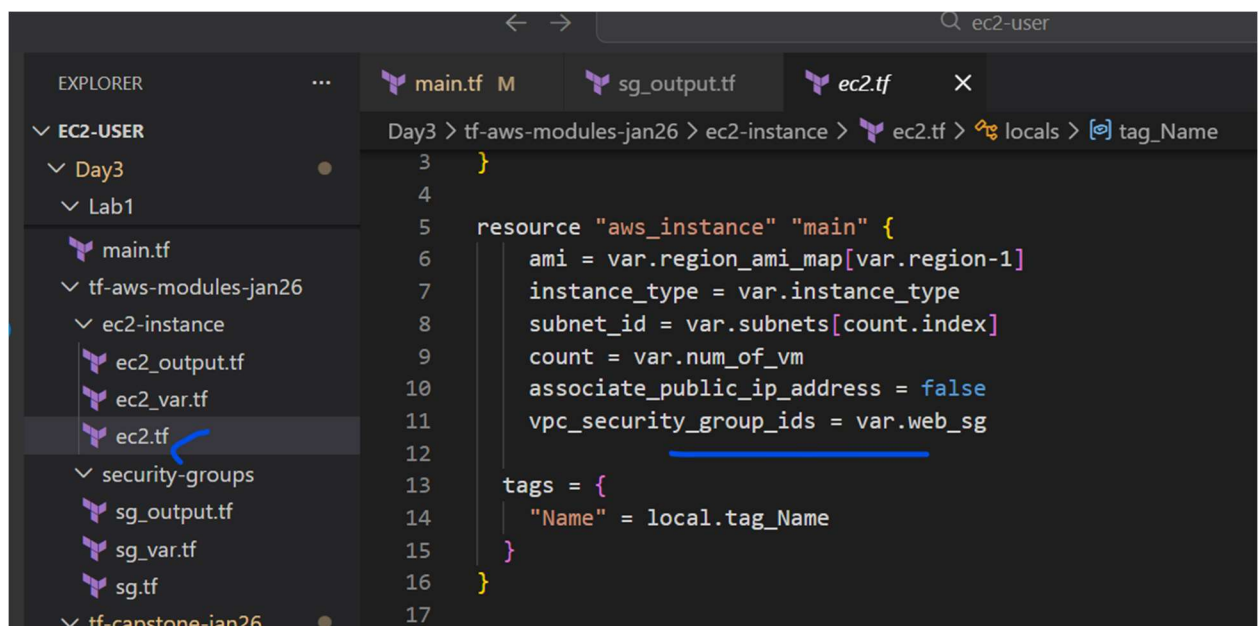
Filename3: sg_output.tf

```

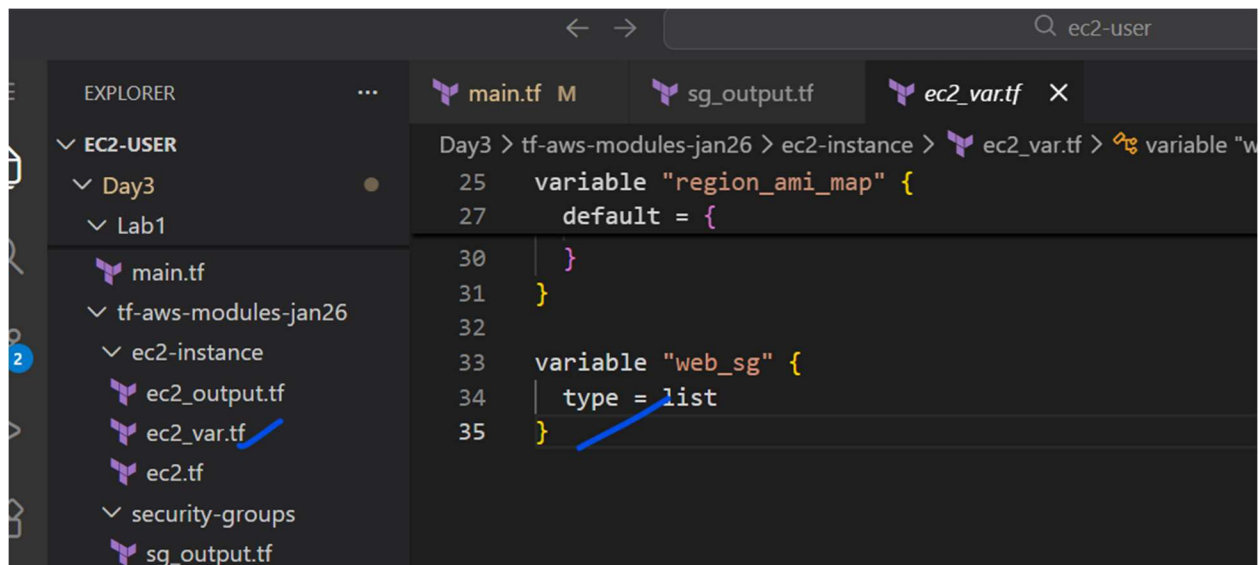
output "web_sg_id" {
    description = "ID of the web security group"
    value       = aws_security_group.web_sg.id
}

```

2. Add security group argument in the “ec2-instance” module

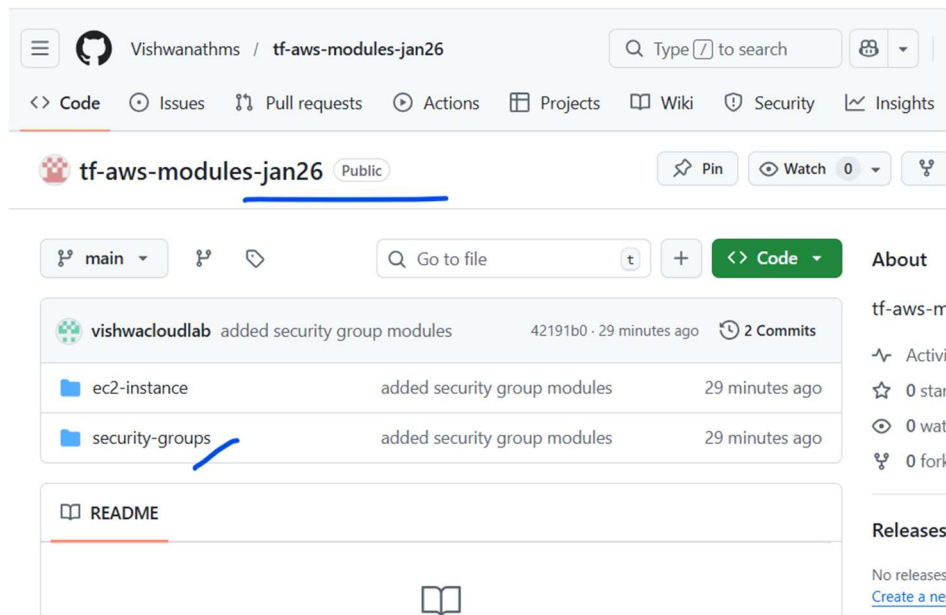


Add the below to the “ec2_var.tf file



```
variable "web_sg" {
  type = list
}
```

3. Push the changes to the module github repo

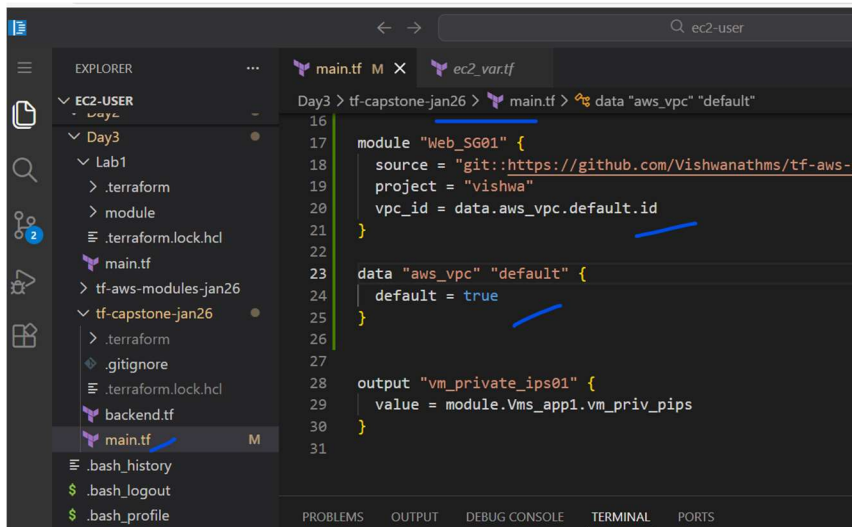


4. Add the module to the main.tf

We are using the data to fetch the VPC ID automatically, instead of hard coding.

```
module "Web_SG01" {
  source = "git::https://github.com/Vishwanathms/tf-aws-modules-
jan26.git//security-groups?ref=main"
  project = "vishwa"
  vpc_id = data.aws_vpc.default.id
}

data "aws_vpc" "default" {
  default = true
}
```



5. Run terraform init and plan

We will get the below error

```
[ec2-user@ip-172-31-18-53 tf-capstone-jan26]$ terraform plan
```

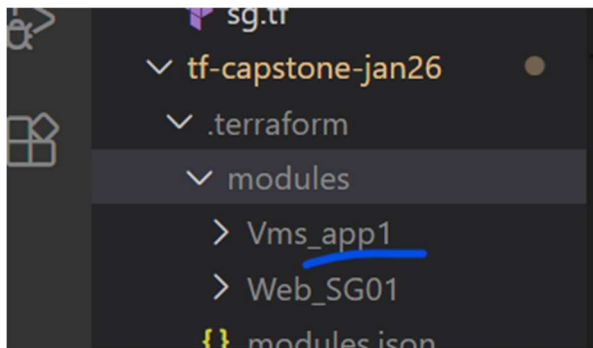
Error: Unsupported argument

on main.tf line 12, in module "Vms_app1":

12: web_sg = [module.Web_SG01.web_sg_id]

An argument named "web_sg" is not expected here.

This is bcz the cached module folder in the .terraform in the capstone has the old module data for "Vms_app1"



Delete that folder "Vms_app1" and run

Terraform init

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

[ec2-user@ip-172-31-18-53 tf-capstone-jan26]$ terraform plan
Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exact
actions if you run "terraform apply" now.
[ec2-user@ip-172-31-18-53 tf-capstone-jan26]$ terraform init
Initializing the backend...
Initializing modules...
Downloading git::https://github.com/Vishwanathms/tf-aws-modules-jan26.git?ref=main for Vms_app1.
- Vms_app1 in .terraform/modules/Vms_app1/ec2-instance
Downloading git::https://github.com/Vishwanathms/tf-aws-modules-jan26.git?ref=main for Web_SG01.
- Web_SG01 in .terraform/modules/Web_SG01/security-groups
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.28.0

Terraform has been successfully initialized!
```

And then the terraform plan and apply will work

Output.

id
ual View 

is

Types
emplates
uests
lans
Instances
Hosts
Reservations
Manager [New](#)

630bd8d0d5cbf

Operator
-

Details | Status and alarms | Monitoring | **Security** | Networking | Storage | Tags

▼ Security details

IAM Role
-

Owner ID
 992382386705

Launch time
Wed Jan 21 2026 14:29:06 GM

Security groups

 sg-0a8b3b7c56646aa20 ([lab9-web-sg-vishwa](#))

▼ Inbound rules

 *Filter rules*

Name	Security group rule ID	Port range	Protocol	Source
lab9-http-rule-vishwa	sgr-05a52216833c05422	22	TCP	0.0.0.0/0
lab9-http-rule-vishwa	sgr-0615f1645c56085f2	80	TCP	0.0.0.0/0